MILITARY SPECIFICATION
INITIATOR, PROPELLANT ACTUATED; JAU-2/A25
METAL PARTS ASSEMBLY

1. SCOPE

1.1 This specification covers Initiator, Propellant Actuated, JAU-2/A25, Metal Parts Assembly. The assembly will ultimately be loaded and used in conjunction with other propellant actuated devices in aircrew emergency escape systems.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposal form a part of this specification to the extent specified herein:

SPECIFICATIONS

Military

MIL-A-2550	- Ammunition and Special Weapons, General Specification for
MIL-I-50375	- Initiator, Propellant Actuated, M104, Metal Parts Assembly

STANDARDS

Military

MIL-STD-105	- Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-109	- Quality Assurance Terms and Definitions
MS-28741	- Hose Assembly, Detachable End Fitting, Medium Pressure
MIL-STD-413	- Visual Inspection Guide for Rubber O-Rings

DRAWINGS

Munitions Command

D11738406	- Initiator, Propellant Actuated, JAU-2/A25, Metal Parts Assembly
D10551447	- Initiator, Propellant Actuated, Metal Parts Assembly

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T20:34Z
Check the source to verify that this is the current version before use.

2.1 DRAWINGS (Cont'd)

C9252573	- Carton, Packing, Ammunition for Initiator, M113
C9252574	- Box, Packing, Ammunition for Initiator, M113
IEL 11738406	- Inspection Equipment List for Initiator, Propellant Actuated, JAU-2/A25, Metal Parts Assembly
IEL 10551447	- Inspection Equipment List, Initiator, Propellant Actuated, Metal Parts Assembly

(Copies of specifications, standards and drawings required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications.- The following document forms a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bid or request for proposal shall apply.

ASA B46.1 - Surface Texture

(Application for copies should be addressed to the American Standards Association, 10 East 40th Street, New York, N.Y. 10016).

3. REQUIREMENTS

3.1 General.- Materials, parts and assemblies shall comply with the requirements specified on the applicable drawings and referenced specifications.

3.2 Shear pin strength.- The shear pin, after application of protective finish, shall shear when a load of 41 pounds plus or minus 5 pounds is applied at the steady rate of 0.125 inches per minute.

3.3 Protective finish.- Protective finish shall be in accordance with the applicable specifications and drawings and shall apply to all surfaces of the component regardless of configuration.

3.4 Residual magnetism.- The metal parts assemblies shall not deflect the indicator of the compass more than 3 degrees in either direction when passed in front of the compass at a distance of 6 inches.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T20:34Z
Check the source to verify that this is the current version before use.

3.5 Workmanship.- The requirements for workmanship are as specified on the applicable drawings, referenced specifications and the following:

3.5.1 Metal defects.- The part shall be free from cracks, splits, cold shuts, inclusions, porosity or other metal defects.

3.5.2 Burr.- The part shall not have a burr which might interfere with the assembly or function of the item or which might be injurious to personnel handling the item.

3.5.3 Foreign matter.- The part or assembly shall not contain chips, dirt, grease, oil, rust, corrosion or other foreign matter.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection.- Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Quality assurance terms and definitions.- Reference shall be made to MIL-STD-109 to define the quality assurance terms used.

4.1.2 Inspection.- Inspection shall be in accordance with MIL-A-2550.

4.2 First article sample.

4.2.1 Initial production sample.- At the beginning of regular production, the contractor shall submit an initial production sample to a government approved facility for evaluation. The sample shall consist of 50 units from each head treated lot for hardness tests, 50 shear pins from each lot, 23 units for protective finish tests, to be tested in accordance with Table I and 10 sets of unassembled parts for visual and dimensional examination.

Source: https://assist.dla.mil -- Downloaded: 202640-22T20:34Z
Check the source to verify that this is the current version before use.

4.2.1 (Cont'd)

Table I

Paragraphs Requirement	Paragraphs Test	Test	Sample Size	Type of Test Destr.	Type of Test Non-Destr.
3.2	4.4.1	Shear Pin Strength	50	X	-
3.3	4.4.2	Protective Finish	In quantities set up in applicable specification	X	-
Dwg.	4.4.3	Hardness	50	X	-
3.4	4.4.4	Residual Magnetism	each	-	X

The sample shall be manufactured using the same materials, equipment, processes and procedures as used in regular production. All parts and materials, including packaging shall be the same as used in regular production and shall be obtained from the same source of supply.

4.2.1.1 Examination and tests.- After examination and provisional acceptance at source, the sample shall be inspected for all requirements of the drawings and specifications at a Government laboratory or such other facility specified in the contract.

4.2.1.2 Initial production failure.- Failure of the sample to comply with the requirements of the drawings and specifications shall result in sample disapproval .

4.3 Inspection provisions.

4.3.1 Lot.

4.3.1.1 Submission of product.- The product shall be submitted in accordance with MIL-STD-105.

4.3.1.2 Size of lot.- A lot shall consist of an identifiable group of metal parts assemblies consisting of components produced by a homogeneous process.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T20:34Z
Check the source to verify that this is the current version before use.

4.3.2 Examination.- One hundred percent examination shall be performed for all critical defects. Examination for major and minor defects shall be performed on a defect basis in accordance with the classification of defects contained herein, using the sampling plans and associated criteria in Tables II and III. For an individual defect (single characteristic), the criteria in Table II shall be used. For total defects within a major or minor group the criteria in Table III shall be used. All non-conforming material shall be rejected. Inspection for rubber O-rings shall be in accordance with MIL-STD-413. The surface roughness comparison specimens prescribed in ASA B46.1 shall be used as a basis of comparison for surface roughness determination. Four power (4X) magnification may be used in performing visual examination for such defects as burr and foreign matter.

Table II

Lot Size	Sample Size	Major		Minor	
		Accept	Reject	Accept	Reject
1 - 300	100% Examination	-	-	-	-
301 - 1300	110	1	2	2	3
1301 - 3200	150	2	3	3	4
3201 - 8000	225	3	4	4	5
8001 - 22000	300	4	5	5	6

Table III

	Total Allowable Defects per Sample Size			
Sample Size for Major Defects	110	150	225	300
No. of Defects in Class				
2 - 3	1	2	3	4
4 - 7	2	3	4	5
8 - 19	3	4	5	7
20 - 43	4	5	8	10
over 43	6	8	11	14
Sample Size for Minor Defects	110	150	225	300
No. of Defects in Class				
2 - 3	2	3	4	5
4 - 7	3	4	5	7
8 - 16	4	5	8	10
17 - 48	6	8	11	14
over 48	8	11	17	20

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T20:34Z
Check the source to verify that this is the current version before use.

4.3.2.1 Pin, Firing, Dwg. C10521573, Covering a detail of Dwg. D11738406

Defect	Method of Inspection
CRITICAL	
1. Diameter of striker	Gage
2. Length of striker	Gage
3. Metal defective	Visual
MAJOR	
101. Concentricity of "O" ring groove with body	Gage
102. Concentricity of striker with body	Gage
103. True position of shear pin hole with body	Gage
104. Diameter of "O" ring groove, min.	Gage
105. Diameter of body, min.	Gage
106. Diameter of shear pin hole, max.	Gage
107. Depth of shear pin hole, max.	Gage
108. Length of "O" ring groove	Gage
109. Location of shear pin hole	Gage
110. Permissible variation of "O" ring groove	Gage
111. Protective finish inadequate or defective	Visual
112. Burr or sharp corner, except where permitted	Visual
113. Foreign matter	Visual
MINOR	
201. Diameter of "O" ring groove, max.	Gage
202. Diameter of body, max.	Gage
203. Diameter of shear pin hole, min.	Gage
204. Depth of shear pin hole, min.	Gage
205. Depth of counterbore	Gage
206. Total length	Gage
207. Length of body to "O" ring groove	Gage
208. Location of marking	Gage
209. Marking missing, incorrect or illegible	Visual
210. Surface finish improper, except as otherwise classified	Visual

4.3.2.2 Housing, Dwg. D11731622, covering a detail of Dwg. D11738406

CRITICAL	
1. Metal defective	Visual

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T20:34Z
Check the source to verify that this is the current version before use.

4.3.2.2 (Cont'd)

Defect	Method of Inspection
MAJOR	
101. Pitch diameter of external thread, min.	Gage
102. Major diameter of external thread, min.	Gage
103. Pitch diameter of internal thread, max.	Gage
104. Minor diameter of internal thread, max.	Gage
105. Concentricity pitch diameter of external thread with pitch diameter of internal thread	Gage
106. Concentricity of center hole with center cavity	Gage
107. Concentricity of cavity at large end with internal thread	Gage
108. Concentricity of internal thread with center cavity	Gage
109. Concentricity of large body with pitch diameter of internal thread	Gage
110. Perpendicularity of base of middle cavity with wall of middle cavity	Gage
111. Perpendicularity of base cavity at large end with pitch diameter of internal thread	Gage
112. Symmetry of shear pin hole with center cavity	Gage
113. Symmetry across flats with internal thread	Gage
114. Diameter of shear pin shaft, max.	Gage
115. Diameter at top of shear pin hole	Gage
116. Diameter of center hole, max.	Gage
117. Diameter of counterbore, large end, max.	Gage
118. Diameter of shoulder	Gage
119. Depth of shear pin hole	Gage
120. Length of internal thread, min.	Gage
121. Length of internal thread, max.	Gage
122. Width of thread undercut	Gage
123. Location of top end of shear pin hole	Gage
124. Location of shear pin shaft	Gage
125. Protective finish inadequate or defective	Visual
126. Surface finish of center cavity, chamfer at mouth of large end and sidewalls of center cavity	Visual
127. Burr or sharp corner	Visual
128. Foreign matter	Visual

Source: https://assist.dla.mil -- Downloaded: 202610-22T20:34Z
Check the source to verify that this is the currentversion before use.

4.3.2.2 (Cont'd)

Defect		Method of Inspection
MINOR		
201.	Pitch diameter of internal thread, min.	Gage
202.	Pitch diameter of external thread, max.	Gage
203.	Diameter of center hole, min.	Gage
204.	Diameter of flange, small end	Gage
205.	Diameter of center cavity	Gage
206.	Diameter of cavity large end	Gage
207.	Diameter of counterbore, large end, min.	Gage
208.	Diameter of chamfer, large end	Gage
209.	Diameter of body, large end	Gage
210.	Diameter of shear pin shaft, min.	Gage
211.	Diameter across flats	Gage
212.	Depth to chamfer of end cavity	Gage
213.	Depth to bottom of cavity, large end	Gage
214.	Depth of counterbore at chamfer end	Gage
215.	Total length	Gage
216.	Length of flats	Gage
217.	Length of hole between cavities	Gage
218.	Thickness of flange	Gage
219.	Thickness of shoulder	Gage
220.	Location of internal thread, max.	Gage
221.	Location of internal thread, min.	Gage
222.	Surface finish improper, except as otherwise classified	Visual

4.3.2.3 Pin, Shear, Dwg. B11731626, covering a detail of Dwg. D11738406

CRITICAL		
1.	Metal defective	Visual
MAJOR		
101.	Diameter, min.	Gage
102.	Total length, min.	Gage
103.	Surface finish improper	Visual
104.	Protective finish inadequate or defective	Visual
105.	Burr	Visual
106.	Foreign matter	Visual

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T20:34Z
Check the source to verify that this is the current version before use.

4.3.2.3 (Cont'd)

Defect	Method of Inspection
MINOR	
201. Diameter, max.	Gage
202. Total length, max.	Gage

4.3.2.4 Bracket, Mounting, D11731741 covering a detail of Dwg. D11738406

CRITICAL	
None defined	
MAJOR	
101. Pitch diameter of internal thread, max.	Gage
102. Minor diameter of internal thread, max.	Gage
103. True position of boss with pitch diameter of internal thread	Gage
104. True position of mounting holes with pitch diameter of internal thread	Gage
105. Diameter of thread lead	Gage
106. Length of effective thread, min.	Gage
107. Length of thread, max.	Gage
108. Metal defective	Visual
109. Protective finish inadequate or defective	Visual
110. Burr	Visual
111. Foreign matter	Visual
MINOR	
201. Pitch diameter of mounting holes	Gage
202. Diameter of mounting holes	Gage
203. Length across open end of bracket	Gage
204. Length across front end of bracket	Gage
205. Total length	Gage
206. Location of mounting holes	Gage
207. Location of large hole	Gage
208. Height of shoulder	Gage
209. Total height including boss	Gage
210. Surface finish improper	Visual

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T20:34Z
Check the source to verify that this is the current version before use.

4.3.2.5 Primer Retainer, Dwg. C11738408, covering a detail of Dwg. D11738406

Defect	Method of Inspection
CRITICAL	
1. Diameter of primer pocket	Gage
2. Depth of primer pocket	Gage
3. Metal defective	Visual
MAJOR	
101. Thickness of wall between primer pocket and relief cavity	Gage
102. Protective finish inadequate or defective	Visual
103. Surface finish of primer pocket, improper	Visual
104. Burr or sharp corner	Visual
105. Foreign matter	Visual
MINOR	
201. Diameter across chamfer	Gage
202. Diameter of relief cavity	Gage
203. Diameter of body	Gage
204. Total length	Gage
205. Radius or chamfer missing or incorrect	Visual
206. Surface finish improper, except as otherwise classified	Visual

4.3.2.6 Assembly, initiator, propellant actuated, JAU-2/A25 - Metal Parts, Dwg. D11738406

CRITICAL	
1. Firing pin protrusion	Gage
2. Metal defective	Visual
MAJOR	
101. Any part missing	Visual
102. Improper assembly	Visual
103. Thread damaged or not full	Visual
104. Burr	Visual
105. Foreign matter	Visual

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T20:34Z
Check the source to verify that this is the current version before use.

4.3.2.6 (Cont'd)

Defect	Method of Inspection
MINOR	
201. "O" ring lubricant missing or inadequate	Visual
202. Marking missing, illegible or incorrect	Visual
203. Varnish or marking missing or inadequate	Visual

4.3.2.7 Carton, packing, ammunition for initiator, Dwg. C9252573

CRITICAL
None defined

MAJOR
None defined

MINOR	
201. Gap between body and cover	Manual
202. Contents loose	Manual
203. Metal ends loose or distorted	Manual
204. Cut or damaged through all impregnated layers	Visual
205. Marking missing, incorrect or illegible	Visual

4.3.2.8 Box, packing, ammunition for initiator, Dwg. C9252574

CRITICAL
None defined

MAJOR
None defined

MINOR	
201. Contents loose	Manual
202. Hardware missing, loose, broken or improperly engaged	Manual and Visual
203. Strapping missing, loose, broken or improperly assembled	Manual and Visual
204. Contents of box exposed	Visual
205. Box damaged	Visual
206. Marking missing, incorrect or illegible	Visual
207. Number of units incorrect	Visual

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T20:34Z
Check the source to verify that this is the current version before use.

4.3.3 Inspection testing.- The following tests shall be performed in accordance with the provisions of 4.4.

4.3.3.1 Shear pin strength.- Fifty pins from each lot of shear pins tested by the contractor shall be tested for shear strength. Failure of any pin to comply with the requirements of 3.2 shall be cause for rejection of the lot.

4.3.3.2 Protective finish.- The number of units for protective finish tests shall be in accordance with the applicable specifications. Failure of any sample to comply with the requirements of 3.3 and the applicable specification shall be cause for rejection of the lot.

4.3.3.3 Hardness.- Fifty units from each heat-treated lot of components shall be tested for the values noted on the applicable drawing. Failure of any unit to comply with drawing requirements shall be cause for rejection of the lot of components.

4.3.3.4 Residual magnetism.- Each assembly shall be subjected to the test for excessive residual magentism. Failure of any assembly to comply with the requirement of 3.4 shall be cause for rejection of the assembly.

4.3.4 Inspection equipment.- The examination and tests shall be made using equipment listed on IEL 11738406 and IEL 10551447.

4.4 Test methods and procedures.

4.4.1 Shear pin strength.- After application of protective finish, the pin shall be subjected to the shear strength test to determine compliance with 3.2. The test shall be performed in the fixture in conjunction with a tensile testing machine.

4.4.2 Protective finish.- Test procedures for protective finish shall be in accordance with the specification referenced on the component drawing.

4.4.3 Hardness.- Prior to the application of protective finish, a hardness tester shall be used to determine compliance with the hardness requirements specified on the applicable drawing.

Source: https://assist.dla.mil -- Downloaded: 202640-22T20:34Z
Check the source to verify that this is the currentversion before use.

4.4.4 Residual magnetism.- The initiator shall be tested for excessive residual magnetism in an area free of local magnetic effects by placing the initiator 6 inches from and at the same height as the compass, in a north-south horizontal position (cap end south) with the compass set in an east-west heading. The initiator shall be moved slowly past the compass in a longitudinal direction for its entire length, rotated 90 degrees in the horizontal pasition about its longitudinal axis and moved slowly back past the compass. This procedure shall be repeated for each 90 degree rotation of the initiator.

5. PREPARATION FOR DELIVERY

5.1 Packing and marking.- Packing and marking shall be in accordance with Drawings C9252573 and C9252574.

6. NOTES

6.1 Ordering data.- Procurement documents should specify the title, number and date of this specification.

6.2 A mechanical failure is defined as any deformation or breakage of a part, the occurrence of which is other than a design function.

Custodian:

Army - MU

Preparing Activity:

Army - MU (FA)

Project No. 1377-A421

Source: https://assist.dla.mil -- Downloaded: 202610-22T20:34Z
Check the source to verify that this is the current version before use.